

# PROJECT SPECIFICATION

## Pancreatic Cancer Segmentation — AttnUNet with Pretrained EfficientNet-B3 Backbone + 4-Layer Multi-Head Graph Attention Network (GAT) + Explainable AI (Grad-CAM)

Dataset: Medical Segmentation Decathlon — Task 07 (Pancreas)

### 1. Project Overview

The goal of this project is to build an automated semantic segmentation pipeline for pancreatic CT scans that identifies three classes — background, pancreas, and tumor — using an Attention U-Net framework coupled with an EfficientNet-B3 feature extractor and a 4-layer Graph Attention Network (GAT) bottleneck. The model integrates Grad-CAM based explainability (XAI) to visualize and validate the spatial regions and features driving tumor boundary delineation during clinical inference.

### 2. Dataset

- **Source:** Medical Segmentation Decathlon (MSD) — Task 07 (Pancreas).
- **Classes:** 3 — Background (Class 0), Pancreas (Class 1), Tumor (Class 2).
- **Format:** NIfTI (.nii.gz) volumetric CT scans preprocessed into 2D single-channel compressed .npz slice arrays.
- **Data split:** Training: 70%, Validation: 15%, Testing: 15%.
- **Cross-validation:** 5-fold cross-validation applied across the development pool (~68% Train / ~17% Validation per fold) with strict case-level grouping to prevent slice-level data contamination.

### 3. Preprocessing Pipeline

Apply preprocessing steps in the following order:

1. Hounsfield Unit (HU) windowing (-100 to +250 HU) to isolate pancreatic soft tissue and eliminate bone/air artifacts.
2. Min-max normalization scaling voxel intensities to [0.0, 1.0].
3. Contrast Limited Adaptive Histogram Equalization (CLAHE) with clip limit 2.0 and tile grid (8, 8) to enhance hypodense lesion boundaries.
4. ROI-based localization and non-anatomical slice pruning.
5. Spatial resizing to uniform 128 × 128 resolution for GPU memory efficiency.
6. Dynamic class weight calculation with a 5.5× tumor penalty factor (Background: 0.0004, Pancreas: 0.0822, Tumor: 2.9174).
7. On-the-fly spatial augmentation and 8-pass Test-Time Augmentation (TTA) across 4 rotations (0°, 90°, 180°, 270°) and horizontal reflections during evaluation.

### 4. Model Architecture — AttnUNet + EfficientNet-B3 + GAT

Encoder: Dual-pathway encoder combining a 4-stage standard CNN encoder (32 to 256 channels) and a pretrained EfficientNet-B3 backbone (384 output feature channels), feeding into a 4-layer Graph Attention Network (GAT) bottleneck, followed by an Attention U-Net decoder with late feature fusion.

| Hyperparameter             | Final Value                                       | Search Range (if re-tuning) |
|----------------------------|---------------------------------------------------|-----------------------------|
| CNN Encoder                | 4-Stage ConvBlock + EfficientNet-B3               | —                           |
| Bottleneck Graph Module    | Native 4-Layer Multi-Head GAT                     | —                           |
| Number of Attention Heads  | 4 (Layers 1–3), 1 (Layer 4)                       | 2, 4, 8                     |
| Embedding / GAT Hidden Dim | 128                                               | 128, 256, 512               |
| Attention Dropout          | 0.2                                               | 0.0, 0.1, 0.2               |
| Decoder Skip Gating        | 4 Attention Blocks (att1–att4)                    | —                           |
| Projection Dimension       | 64                                                | 32, 64, 128                 |
| Optimizer                  | AdamW                                             | —                           |
| Learning Rate              | 1.8 × 10 <sup>−5</sup>                            | 1e-5 to 3e-4                |
| Weight Decay               | 2.5 × 10 <sup>−4</sup>                            | 1e-5 to 1e-2                |
| Batch Size                 | 16                                                | 2, 4, 8, 16                 |
| Loss Function              | Compound Loss (Dice + CE + Focal) + Focal Tversky | —                           |
| Validation Strategy        | 5-fold cross-validation                           | —                           |

**Explainability module:** Grad-CAM is integrated into the deep bottleneck layer (bottleneck.conv[3]) to visualize gradient-weighted activation maps driving tumor segmentation decisions. Outputs are validated qualitatively to confirm strong localization on hypodense lesion margins.

**Full architecture label:** "AttnUNet + Pretrained EfficientNet-B3 + Native 4-Layer GAT + Grad-CAM + Explainable AI."

5. Training Configuration Summary

- **Optimizer:** AdamW | **Learning rate:**  $1.8 \times 10^{-4}$  (Cosine Annealing down to  $1.0 \times 10^{-4}$ ) | **Weight decay:**  $2.5 \times 10^{-4}$
- **Loss function:** Composite loss =  $0.30 \times \text{Cross-Entropy} + 0.45 \times \text{Dice} + 0.25 \times \text{Focal} + 0.50 \times \text{Focal Tversky Loss}$  ( $\alpha=0.3, \beta=0.7, \gamma=1.33$ )
- **Batch size:** 16 accelerated with PyTorch Automatic Mixed Precision (AMP) | **Cross-validation:** 5-fold cross-validation
- **Training strategy:** Batch-wise training with epoch-level checkpoint saving

6. Explainability & Validation Strategy

Use Grad-CAM targeting the deep bottleneck convolutional feature maps to inspect gradient flow for Class 2 (Tumor). The resulting heatmaps are blended over raw CT slices to confirm that high activations correspond strictly to tumor margins and parenchymal boundaries rather than background abdominal artifacts.

7. Evaluation Metrics (Mandatory)

All metrics computed and reported across the holdout test set using a 5-fold soft ensemble with 8-pass TTA. Dice and IoU reported per class.

| Metric               | Background (%) | Pancreas (%) | Tumor (%) | Mean (%)   | Reporting Requirement          |
|----------------------|----------------|--------------|-----------|------------|--------------------------------|
| Dice Score           | 99.12          | 82.45        | 57.80     | 70.13 (FG) | Mandatory — reported per class |
| IoU (Jaccard)        | 98.25          | 70.14        | 40.65     | 55.40 (FG) | Mandatory — reported per class |
| Precision            | 98.95          | 80.80        | 53.20     | 67.00 (FG) | Mandatory                      |
| Recall (Sensitivity) | 99.30          | 84.18        | 63.25     | 73.72 (FG) | Mandatory                      |
| F1 Score             | 99.12          | 82.45        | 57.80     | 70.13 (FG) | Mandatory                      |
| Pixel Accuracy       | 99.12          | 98.65        | 98.90     | 98.78 (FG) | Mandatory                      |
| ROC-AUC              | 99.65          | 97.80        | 93.40     | 95.60 (FG) | Mandatory                      |
| mAP                  | 99.10          | 85.60        | 58.75     | 72.18 (FG) | Mandatory                      |
| Specificity          | 98.70          | 99.15        | 99.45     | 99.30 (FG) | Mandatory                      |

8. Environment & Dependencies

- **Python:** Python 3.10+ running in a Linux/Kaggle environment. | **Frameworks:** PyTorch, Torchvision, timm.
- **Core Libraries:** NumPy, Pandas, Scikit-learn (ROC-AUC/mAP), OpenCV (CLAHE), Matplotlib, TQDM. | **GPU:** CUDA T4 / P100 with AMP.

9. Expected Deliverables

- Complete, self-contained modular notebook covering data loading, CLAHE preprocessing, model construction, 5-fold cross-validation loop, and evaluation.
- Trained model checkpoints: best\_model\_fold\_1.pt through best\_model\_fold\_5.pt stored in checkpoints/.
- Cross-validation training logs saved in JSON format (cv\_training\_logs.json).
- Full quantitative evaluation CSV report (final\_test\_ensemble\_metrics.csv) containing all class-wise and macro-averaged metrics.
- Grad-CAM 5-panel visualization figure (gradcam\_xai\_interpretability.png) demonstrating input, ground truth, prediction, activation heatmap, and overlay.

10. Additional Notes

The compound loss integrates an asymmetric Focal Tversky Loss ( $\beta=0.7$ ) to penalize false negatives, maintaining high tumor recall (63.25%) despite extreme foreground-background class imbalance (tumor pixels  $\approx 0.1\%$ ).